INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

DEPARTMENT OF PHYSICS

END-AUTUMN SEMESTER EXAMINATION 2024-25

Subject: **Classical Mechanics (PHPH31207)** Total Marks: **50**

Time Duration: **3 Hours**

No. of students: **99** Date: **22-11-2024** & Time: **2PM- 5PM**

Answer all the questions (Start each question on a fresh page)
(If the given information is not sufficient make proper assumptions and mention them clearly)
(Take g = 10 m/s^2, e = 2.72 where ever it is necessary)

1. Find the equation of motion and force constraint in the case of a simple pendulum (Fig. 1) using the method of Lagrange's undetermined multipliers. [6M]
2. (a) Find the position (q) and momentum (p) coordinates for a harmonic oscillator. The harmonic oscillator is described by the Hamiltonian $H = \frac{1}{2}(p^2 + \omega^2 q^2)$, and generating function $F = \frac{1}{2}\omega q^2 \cot(2\pi Q)$. [4M]
   (b) Show that the transformation $P = q\cot p$, $Q = ln\left(\frac{\sin p}{q}\right)$ is canonical and find the generating function of the form F (q, Q). [4M]
3. If $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $\vec{p} = p_x\hat{\imath} + p_y\hat{\jmath} + p_z\hat{k}$, find the value of the Poisson bracket $[|\vec{r}|, |\vec{p}|]$. [4M]
4. Show that the Poisson bracket is invariant under canonical transformation. [8M]
5. (a) If the following transformation equations $Q = q^m \cos np$, $P = q^m \sin np$ represent canonical transformations, find the values of m and n using Poisson brackets. [4M]
   (b) For the above transformation, find the generating function of the form F (p, Q). [2M]
6. A projectile is projected with a speed 'u' at an angle 'θ' with the horizontal. Using the Hamilton-Jacobi method, show that the path of the projectile is a parabola and obtain its equations [The motion of the projectile is in the (x, y) plane with y as the vertical axis]. [8M]
7. Obtain the frequency of the harmonic oscillator described by the Hamiltonian $H = \frac{p^2}{2m} + \frac{kq^2}{2}$ using Action-Angle variable method. [4M]
8. State and prove Liouville's theorem (Consider the case of a mechanical system with one degree of freedom). [6M]

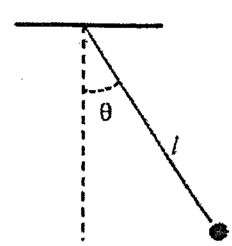

Fig. 1